

Zhenbang Zhang

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

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Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Sept. 2025 – Present

M.Sc. in Machine Learning, GPA: 3.7/4.0

Abu Dhabi, UAE

Shandong University

Sept. 2021 – June 2025

B.S. in Computer Science, GPA: 3.52/4.00

Qingdao, China

Research Interests

Computational connectomics; synaptic-resolution connectome reconstruction; dense neuron segmentation and tracking; machine learning for connectomics dataset reconstruction, annotation, and analysis

Publications

- **Zhang, Z.[†]**, Li, H.[†], Yang, Z., Xu, Z., Sun, D., Gao, X., Zhang, F., & Han, R. (2026). vEMRec: High-resolution volume electron microscopy reconstruction based on structure-preserving and high-fidelity 3D alignment. *Advanced Science*.
- **Zhang, Z.[†]**, Li, H.[†], Feng, J., El-Messiry, H., Xu, Z., & Han, R. (2026). 3D Registration-Guided Deformable Residual Inpainting for ssEM Restoration. *Bioinformatics* (accepted).
- **Zhang, Z.[†]**, Cui, Z.[†], El-Messiry, H., Han, R., & Xu, Z. (2026). Motion-Residual Conflict-Aware Time Reversal for Generative Inbetweening. In *Proceedings of the Forty-Third International Conference on Machine Learning*. (accepted).
- **Zhang, Z.[†]**, Feng, J.[†], Li, H., El-Messiry, H., Xu, Z., & Han, R. (2025). Unsupervised trajectory optimization for 3D registration in serial section electron microscopy using Neural ODEs. In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*.
- **Zhang, Z.[†]**, Li, H.[†], Xu, Z., Meng, W., & Han, R. (2025). A Gaussian filter-based 3D registration method for serial section electron microscopy. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 39, No. 1, pp. 1156–1164)*.
- He, B.[†], Zhang, Y.[†], **Zhang, Z.[†]**, Cheng, Y., Zhang, F., Sun, F., & Han, R. (2024). vEMstitch: An algorithm for fully automatic image stitching of volume electron microscopy. *GigaScience*, 13, giae076.

[†] Equal contribution.

Research Experience

Research Assistant

Oct. 2025 – Present

MBZUAI, Advisors: Prof. Zhiqiang Xu

Abu Dhabi, UAE

- Lead research on **isotropic reconstruction and image restoration** for volume EM connectomics, improving z-axis resolution, recovering degraded ultrastructure, and enhancing 3D volume quality for downstream synaptic-resolution reconstruction; resulting in **first-author papers at ICML 2026 and Bioinformatics**.
- Investigate **automatic proofreading** for large-scale connectome reconstruction, aiming to correct

segmentation errors, reduce manual proofreading burden, and improve the reliability of analysis-ready connectomics datasets.

Research Intern

Jul. 2023 – Oct. 2025

Mathematics and Interdisciplinary Science Center, Shandong University, Advisor:
Prof. Renmin Han

Qingdao, China

- Developed a Gaussian filter-based 3D registration method for serial-section EM alignment, introducing a problem-reformulation perspective beyond conventional pairwise registration and achieving **about 30% improvement in accuracy**; resulting in a **first-author paper at AAAI 2025**.
- Proposed an unsupervised trajectory-optimization framework based on Neural ODEs, reframing serial-section EM alignment as continuous trajectory optimization to improve cross-section structural consistency; resulting in a **first-author paper at NeurIPS 2025**.
- Developed *vEMRec*, an end-to-end 3D alignment pipeline for volume electron microscopy, and validated it on six real-world datasets exceeding **1 TB**, improving ultrastructural recovery and processing efficiency for connectome-scale analysis; resulting in a **first-author paper in Advanced Science**.

Research Intern

Jun. 2022 – Jul. 2023

Mathematics and Interdisciplinary Science Center, Shandong University, Advisor:
Prof. Renmin Han

Qingdao, China

- Developed **automatic stitching** methods for ultra-high-resolution vEM image sequences, supporting large-scale 3D reconstruction and downstream connectomics workflows.
- Co-first-authored *vEMstitch*, a fully automatic stitching method for volume electron microscopy; resulting in a **co-first-author paper in GigaScience**.

Teaching Experience

Teaching Assistant, Course: Linear Algebra

2023 – 2024

Shandong University

Qingdao, China

- Supported undergraduate instruction in linear algebra through problem-solving sessions, office hours, and assignment grading.
- Strengthened scientific communication and mentoring skills through one-on-one support.

Skills

Programming: Python, C++, C, MATLAB, LaTeX

Languages: Chinese (Native), English (IELTS: 6.5)